Jiaqi (Jake) Zhao

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Research Interests

Understanding the physical nature, emission mechanisms, and variability of compact objects, including neutron stars, white dwarfs, and compact binaries; exploring unusual sources and transient phenomena, such as millisecond pulsars, magnetars, long-period radio transients, and fast radio bursts; and investigating their underlying physics through multiwavelength observations, with an emphasis on X-rays complemented by radio and optical data.

Education

- **Ph.D. in Astrophysics**, **Sept. 2021–Aug. 2025**
  University of Alberta, Canada
  Thesis Title: "The Fascinating Neutron Stars in X-rays: Millisecond Pulsars, Globular Clusters, and Beyond"
  Supervisor: Prof. Craig O. Heinke
- **M.Sc. in Physics**, **Sept. 2019–Sept. 2021**
  University of Alberta, Canada
  Thesis Title: "X-ray Millisecond Pulsars in the Galactic Globular Cluster"
  Supervisor: Prof. Craig O. Heinke
- **B.Sc. in Physics**, **Sept. 2014–Jun. 2018**
  Southwest University, China
  Advisor: Prof. Xiaopeng You

Employment

- **Shuimu Postdoctoral Fellow** October 2025 – Present
  Department of Astronomy, Tsinghua University Beijing, China
  Investigate compact objects and transient phenomena through X-ray and radio observations. Coordinate a high-speed optical sampling backend project for a 1.2-m telescope.
- **Teaching Assistant** September 2019 – December 2021
  Department of Physics, University of Alberta Edmonton, AB, Canada
  Supported undergraduate teaching in *Physics of Energy*, *Modern Physics*, and *Galactic & Extragalactic Astrophysics*, among other courses.

Publications

First-Author Publications (5)

1. **Zhao, J.**, Heinke, C. O., Shternin, P. S., Ho, W. C., Ofengeim, D. D., & Patnaude, D. (2025). Verification of Cas A neutron star cooling rate using Chandra HRC-S observations. *The Astrophysical Journal*, 991(2), 136. doi: 10.3847/1538-4357/adfdcf. arXiv: 2508.15161.
2. **Zhao, J.**, Heinke, C. O., & Fu, S. (2025). A deep Chandra study confirms diffuse nonthermal X-ray emission from the globular cluster Terzan 5. *Astronomy & Astrophysics*, 693, A107. doi: 10.1051/0004-6361/202451217. arXiv: 2412.03788.
3. **Zhao, J.**, & Heinke, C. O. (2023). A Chandra X-ray study of millisecond pulsars in the globular cluster Omega Centauri: a correlation between spider pulsar companion mass and X-ray luminosity. *Monthly Notices of the Royal Astronomical Society*, 526(2), 2736–2753. doi: 10.1093/mnras/stad2930. arXiv: 2309.13189.

4. **Zhao, J.**, & Heinke, C. O. (2022). A census of X-ray millisecond pulsars in globular clusters. *Monthly Notices of the Royal Astronomical Society*, 511(4), 5964–5983. doi: 10.1093/mnras/stac442. arXiv: 2202.07040.
5. **Zhao, J.**, Zhao, Y., & Heinke, C. O. (2021). Chandra and HST studies of six millisecond pulsars in the globular cluster M13. *Monthly Notices of the Royal Astronomical Society*, 502(2), 1596–1604. doi: 10.1093/mnras/stab117. arXiv: 2101.07986.

Co-Authored Publications (7)

1. Zhang, L., Fu, X., Li, D., Carli, E., Wang, L., **Zhao, J.**, Heinke, C. O., et al. (2026). A pulsar escaping an ancient open cluster via tidal stripping. *Science Bulletin*, 71(16), 4103–4106. doi: 10.1016/j.scib.2026.07.006. arXiv: 2606.27796.
2. Li, D., Yuan, M., Wu, L., Yan, J., Lv, X., Tsai, C., Wang, P., **Zhao, J.**, Zhu, W., . . . , Wang, C. (2026). A 44-min periodic radio transient in a supernova remnant. *Science Bulletin*, 71(14), 3609–3616. doi: 10.1016/j.scib.2026.06.015.
3. Hebbar, P. R., **Zhao, J.**, Heinke, C. O., & Mehdwan, J. K. (2026). Using Fe K-line emission to separate X-ray binary and millisecond pulsar candidates from cataclysmic variables in the Galactic bulge. *The Astrophysical Journal*, 1001(1), 100. doi: 10.3847/1538-4357/ae4dee.
4. Kumawat, G., Heinke, C. O., **Zhao, J.**, Bahramian, A., Cohn, H. N., & Lugger, P. M. (2025). A Comprehensive Analysis of X-Ray Sources in Terzan 5 Using Chandra Observations. *The Astrophysical Journal*, 990(2), 218. doi: 10.3847/1538-4357/adf724. arXiv: 2508.02526.
5. Zhao, Y., D' Antona, F., Milone, A. P., Heinke, C., **Zhao, J.**, Lugger, P., & Cohn, H. (2024). Exploration of faint X-ray and radio sources in the massive globular cluster M14: A UV-bright counterpart to Nova Ophiuchus 1938. *Monthly Notices of the Royal Astronomical Society*, 527(4), 11491–11506. doi: 10.1093/mnras/stad3980. arXiv: 2401.02854.
6. Zhang, L., Freire, P. C., Ridolfi, A., Pan, Z., **Zhao, J.**, Heinke, C. O., ... & Li, D. (2023). Discovery and Timing of Millisecond Pulsars in the Globular Cluster M5 with FAST and Arecibo. *The Astrophysical Journal Supplement Series*, 269(2), 56. doi: 10.3847/1538-4365/acfb03. arXiv: 2312.05835.
7. Lugger, P. M., Cohn, H. N., Heinke, C. O., **Zhao, J.**, Zhao, Y., & Anderson, J. (2023). Exotica in the Globular Cluster M4, Studied with Chandra, HST, and the VLA. *Monthly Notices of the Royal Astronomical Society*, doi: 10.1093/mnras/stad1887. arXiv: 2306.11770.

Seminars & Conference Presentations

- "Verification of Cas A neutron star cooling rate using Chandra HRC-S observations" (contributed talk), X-ray Universe 2026, Elche, Spain (June 2026). Also presented as posters at the CASCA Annual General Meeting, Halifax, NS, Canada, and the European Astronomical Society Annual General Meeting, Cork, Ireland (June 2025).
- "Chandra X-ray study of millisecond pulsars in the globular cluster Omega Centauri" (poster), CASCA Annual General Meeting, Penticton, BC, Canada (June 2023)
- "X-ray millisecond pulsars in globular clusters" (30-minute talk), Clusters Workshop at McMaster University, Hamilton, ON, Canada (August 2022)
- "A census of X-ray millisecond pulsars in globular clusters" (poster), CASCA Annual General Meeting, Online (May 2022)
- "Chandra X-ray observations of 6 millisecond pulsars in the globular cluster M13" (poster), CASCA Annual General Meeting, Online (May 2020)

Awarded Observing Time

Principal Investigator

FAST, 3.5 hr 2026
Searching for Radio Pulsations from the Candidate White Dwarf Pulsar SDSS J230641.47+244055.8, Proposal ID: PT2026_0058

FAST, 5 hr 2026
Hunting white dwarf pulsars with FAST: a targeted radio search of short-period white dwarf-M dwarf binaries, Proposal ID: PT2026_0083

FAST, 6.5 hr 2026
Testing Whether Benchmark Long-period Radio Transients Are Truly Silent in Their Inactive States, Proposal ID: PT2026_0140

Co-Investigator

Chandra X-ray Observatory, 200 ks 2026
A deep X-ray survey of the massive globular cluster M14, Proposal ID: 28400116

Grants and Fellowships

- **Shuimu Postdoctoral Fellowship**, Tsinghua University (October 2025–Present)
- **China Scholarship Council Scholarship**, China Scholarship Council (September 2021–August 2025), CAD 26,400 annual stipend plus tuition
- **Hanhong Scholarship**, Hanhong College, Southwest University (2020), CNY 5,000
- **University of Alberta Graduate Fellowship**, University of Alberta (September 2019–August 2020), CAD 8,000

Academic Service

- **Journal Referee**: *The Astrophysical Journal* (ApJ).
- **Thesis Defense Secretary**: Three defenses, Tsinghua University.

Skills

- **X-ray astronomy**: CIAO, Sherpa, DS9, BXA.
- **Radio astronomy (practical experience)**: PRESTO, Riptide.
- **Programming and software**: Python, LaTeX, TOPCAT.

Astronomical Catalogs

- **LPT-Cat**: Catalog of long-period radio transients (*published online*).
- **RAWDOG**: Catalog of radio-emitting white dwarf systems (*published online*).
- **GCPIX**: Catalog of X-ray properties of globular-cluster pulsars (*ongoing*).